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## NumPy ufunc

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# NumPy Differences

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## Differences

A discrete difference means subtracting two successive elements.

E.g. for [1, 2, 3, 4], the discrete difference would be [2-1, 3-2, 4-3] = [1, 1, 1]

To find the discrete difference, use the `diff()` function.

## Example

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Compute discrete difference of the following array:

```
import numpy as np

arr = np.array([10, 15, 25, 5])

newarr = np.diff(arr)

print(newarr)
```

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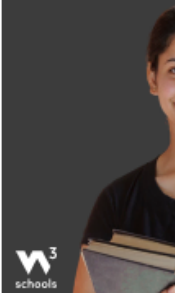
**Returns:** [5 10 -20] because 15-10=5, 25-15=10, and 5-25=-20

We can perform this operation repeatedly by giving parameter `n`.

E.g. for [1, 2, 3, 4], the discrete difference with `n = 2` would be [2-1, 3-2, 4-3] = [1, 1, 1], then, since `n=2`, we will do it once more, with the new result: [1-1, 1-1] = [0, 0]

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## Example

Compute discrete difference of the following array twice:

```
import numpy as np

arr = np.array([10, 15, 25, 5])

newarr = np.diff(arr, n=2)

print(newarr)
```

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**Returns:** `[5 -30]` because:  $15-10=5$ ,  $25-15=10$ , and  $5-25=-20$  AND  $10-5=5$  and  $-20-10=-30$

## Exercise ?

Consider the following code:

```
import numpy as np
arr = np.array([1, 2, 3, 4])
newarr = np.diff(arr)
```

What will be the result of newarr ?

- ☐ [1 1 1 1]
- ☐ [1 1 1]
- ☐ -2

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